

Phase 10: Skeletal character rigs

Future extension of Stagehand's current pose-based character rig

Status

This is a future extension of Stagehand's current pose-based character rig. It is not part of the MVP checkpoint.

Product goal

Let a person provide a character reference image or pose sheet and receive a reviewable skeletal-rig proposal that can be corrected before it drives animation.

Proposed flow

- Import a character reference image or pose sheet and preserve the original asset as the source of truth.
- Generate a proposal containing a root, joints, bones, attachment regions, and confidence per landmark.
- Show the proposal over the reference with clear handles for moving, adding, removing, and mirroring joints.
- Let the person approve or revise the joint graph and art binding before animation commands can use it.
- Create bone transforms and optional mesh or segmented-cutout weights while retaining the existing rigid-pose fallback.
- Animate bones with the same revisioned commands, undo/redo history, playhead, Preview evaluator, validation, and WebM renderer used by the current studio.

Suggested structured model

- Skeleton: stable id, root joint, joints, bones, bind asset, binding method, version, and approval state.
- SkeletonJoint: id, parent id, x/y, radius, label, confidence, and locked state.
- SkeletonBone: id, parent joint, child joint, length, angle limits, and optional influence regions.
- SkeletonBinding: asset id, method (rigid, segmented, or mesh), weights or regions, and source-image dimensions.
- BoneKeyframe: scene id, skeleton id, time, joint or bone transforms, and interpolation mode.

Agent surface

Candidate tools are `propose_skeleton`, `get_skeleton`, `update_skeleton_joint`, `set_bone_keyframe`, `bind_skeleton_asset`, and `validate_skeleton`. Proposal and binding tools must return a pending review state rather than silently making a character animation-ready.

Acceptance gates

- A reference can produce a deterministic, inspectable proposal or an explicit UNABLE_TO_PROPOSE result.
- The person can correct every joint and binding region before approval.
- Unapproved skeletons cannot affect Preview or export.
- Human and agent edits share revision guards, idempotency, undo/redo, persistence, and validation.
- The renderer, thumbnails, Preview, frame inspection, and WebM export agree on the same bone-evaluated frame.
- Existing six-pose characters and pose-sheet imports continue to work unchanged.
- A failed or low-confidence proposal never destroys the original reference or the rigid-pose fallback.

Deliberate non-goals

- Training a custom vision model, automatic perfect weights, 3D motion capture, facial lip-sync, or replacing the current deterministic pose rig.
- Those can be evaluated only after the reviewable 2D skeleton workflow is reliable.